

Steps Before the Answer

DATA 443/643 · Week 13, note 13.1

Week 13

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Ideas from Stanford CS224N 2026 L12 (CoT is extra tokens). Original slides; that course is not copied.

Lecture plan

- 1 A prompt, not a new architecture
- 2 Tokens as scratch paper
- 3 17×24 in three lines
- 4 When extra decode hurts

Key goal: CoT is a decoding choice. Report the prompt and the box.

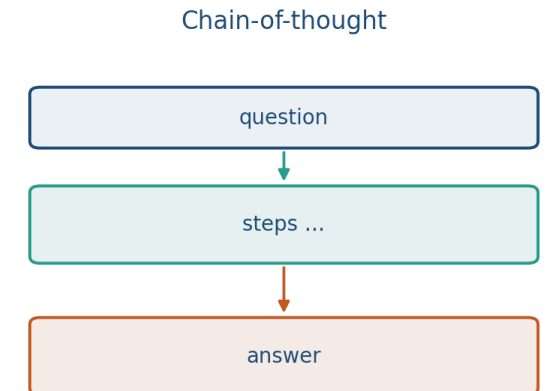
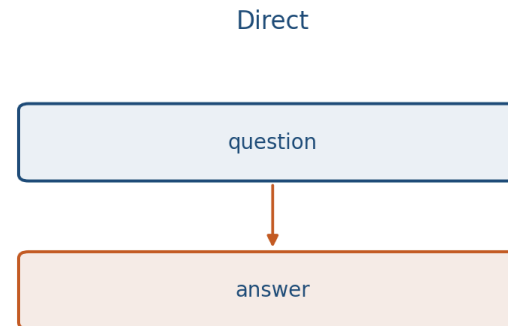
First time: what / why / formula

- What / why. Direct: “What is 17×24 ?” CoT: “Show your work, then the number.” Few-shot: paste worked traces. Zero-shot: *Let’s think step by step.* Extra tokens are extra test-time compute. Multi-hop arithmetic can use the tape. Weights do not change.
- Architecture. Same LM. Longer generation. Parse a final boxed answer separately from the trace (note 13.3).
- Formula. Still $\prod_t p(y_t \mid y_{<t}, \text{prompt})$; the prompt asks for steps. No new parameters.
- Tradeoffs. + Helps multi-hop. – Hurts one-hop lookup; latency/cost; fluent wrong algebra (faithfulness next). Not retrieval or tools.

How. Freeze the instruction and decoding settings. Temperature 0 is one greedy chain, not a vote (13.2).

Ask for work, then the number

- Direct: "What is 17×24 ?" CoT: "Show your work."
- CS224N L12: the weights do not change. The tape gets longer.
- Few-shot traces must match the test task.



Partial products on the tape

$$17 \times 24 = 10 \times 24 + 7 \times 24 = 240 + 168 = 408$$

Three lines hold 240, 168, then 408.

Direct decode has no place to put the 240.

Lab 13 uses the same algebra on $23+19$.

A tape you can inspect

- Gold is 408. This trace is also faithful (13.3).
- A one-hop lookup does not need hops. Extra steps wander.
- Freeze temperature and max tokens in the report.

17 × 24: three lines hold the partial products. Direct decode has no tape.

step	value
10 × 24	240
7 × 24	168
sum	408

When the extra tokens help

Single lookup

Capital of France. A story about Lyon is extra risk, not extra reasoning.

Multi-hop

Dependent arithmetic, a plan, then a box. Intermediate tokens store the hops.

What to take from this lecture

Not a fine-tune

Unless you trained on traces, CoT is a prompt.

Two numbers

Answer accuracy, and a trace check in 13.3.

Lab 13

Constructed strings. No LLM API.